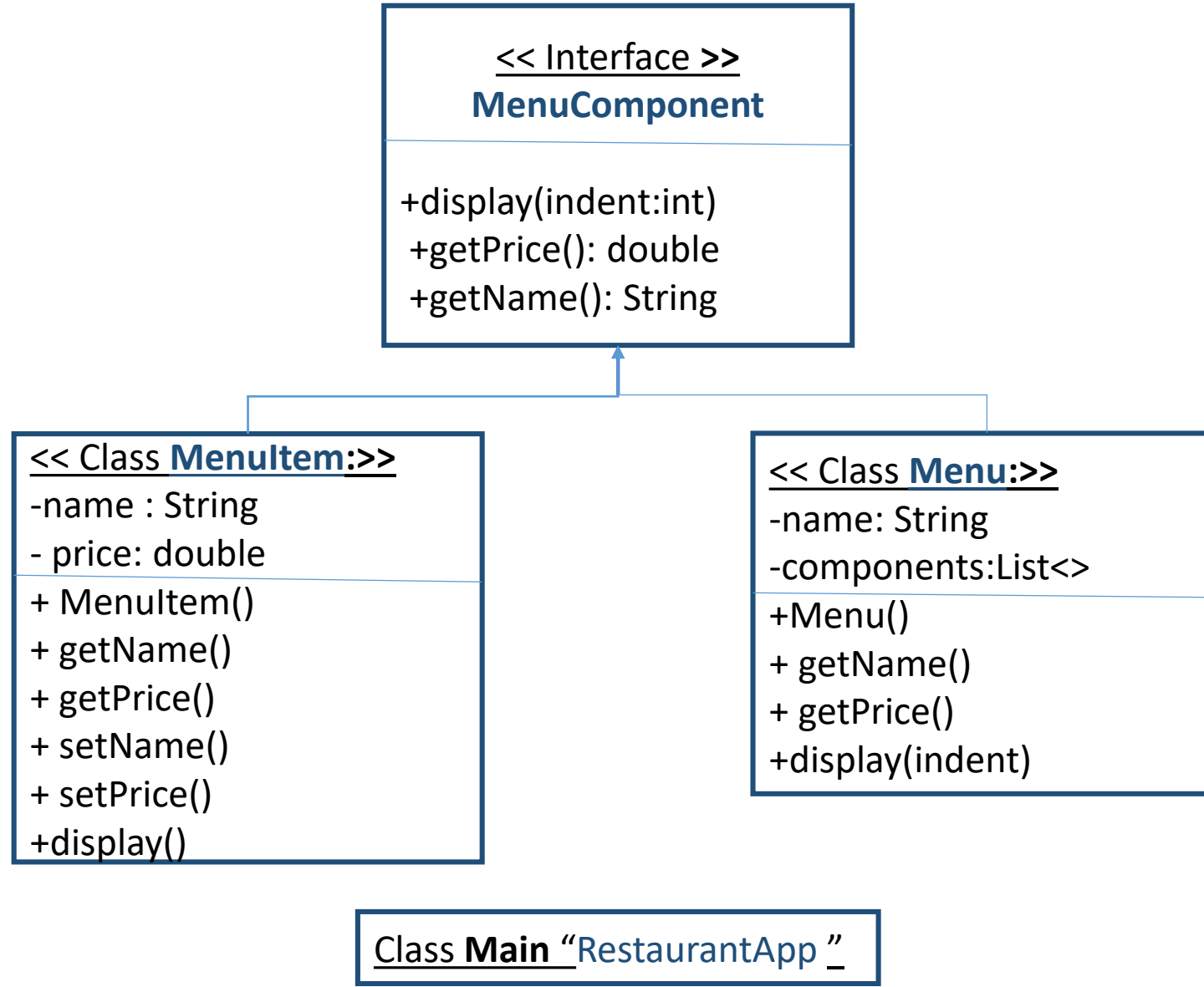




Package “restaurantapp”



Technical Documentation

The Restaurant Menu System is a Java application developed using the Composite Design Pattern.

The system simulates a restaurant menu where:

- * Individual food items can be created.
- * Menus can contain multiple items.
- * Menus can also contain other menus.

This structure provides flexibility and allows the program to treat single objects and groups of objects uniformly.

Objective :

- * Demonstrate the implementation of the Composite Design Pattern in Java.
- * Build a hierarchical restaurant menu system.
- * Allow recursive operations such as displaying menus and calculating total prices.

Design Pattern Used:

Purpose: The Composite Pattern is used to represent hierarchical tree structures.

It allows:

- * Individual objects (MenuItem)
- * Composite objects (Menu)

to be handled using the same interface (MenuComponent).

How the Pattern is Applied:

Pattern Role	Class
Component	MenuComonent
Leaf	MenuItem
Comosite	Menu
Client	RestaurantApp

Class Descriptions:

1. MenuComponent Interface

Method	Description
display(int indent)	Displays menu structure
getPrice()	Returns price
getName()	Returns component name

2. MenuItem Class

Attributes	Type
name	String
price	double

Responsibilities:

- * Store item data.
- * Display item information.
- * Return item price.

3. Menu Class

Attribute	Type
name	String
components	List

Responsibilities:

- * Store menu components.
- * Add items or submenus.
- * Calculate total prices recursively.
- * Display hierarchical menu structure.

4. RestaurantApp Class

Responsibilities:

- * Create menu items.
- * Create menus.
- * Add components to menus.
- * Display final output.

System Workflow

1. Create individual MenuItem objects.
2. Create Menu objects.
3. Add items or menus into menus.
4. Display the complete menu hierarchy.
5. Calculate the total menu price recursively.

Advantages of Using Composite Pattern

- * Simplifies client interaction.
- * Supports recursive structures.
- * Improves maintainability.
- * Allows uniform handling of objects and object groups.
- * Makes the system scalable and flexible.

Conclusion

The Restaurant Menu System successfully demonstrates the Composite Design Pattern in Java.

The implementation allows menus and menu items to be managed using a unified structure while supporting nested menu hierarchies and recursive operations.